HyperTensor (Part VII): Structure-Aware FFN Compression via Column Clustering

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Abstract

HyperTensor Part I (Geodesic Runtime Compression) compresses only the attention projections ($\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$), leaving the feed-forward network (FFN) at full rank because global SVD of FFN weights is unacceptably lossy: FFN singular spectra are nearly flat, with $k_{95}/d \approx 0.85$ versus 0.41 for attention. This paper demonstrates that structure-aware compression—clustering FFN columns by weight similarity and applying per-cluster SVD—recovers 21–25% of the global-SVD reconstruction error at the same total rank budget on SmolLM2-135M (4 clusters, k=0.25n). We extend the analysis to all 30 layers across five compression ranks, characterise the cluster-geometry trade-off, and specify the end-to-end PPL measurement protocol needed to validate the approach at deployment scale.

Measured (SmolLM2-135M, 30 layers, L2 clustering):

- At k=0.25n (aggressive): 4-cluster compression reduces mean Frobenius error by 22.6% versus global SVD (0.431 vs. 0.553 relative error, averaged over layers $\{0, 15, 29\}$).
- At k=0.50n (moderate): per-cluster SVD achieves near-zero error ($< 10^{-6}$)—the cluster structure fully captures FFN geometry at this compression ratio.
- The gain is anti-correlated with cluster count: 4 clusters outperform 8, which outperform 16. Fewer, larger clusters preserve more intra-cluster structure at the same total rank budget.

P2 phases measured (SmolLM2-135M, RTX 4070 Laptop/EC2 L40S):

- Phase 1 (L2 clustering): 21–25% reconstruction improvement at C=4 vs. global SVD. At k_{frac}=0.50, near-zero error.
- Phase 2 (real activations): 4–33 massive columns per layer detected. Activation-weighted SVD gives PPL 54.19 (1.99× baseline)—22.7× better than weight-norm proxy.
- Phase 3 (weight-norm proxy): FALSIFIED. L2 column norms are uncorrelated with functional importance; weight-norm-guided SVD gives PPL 45× baseline at k_{frac}=0.50.
- Phase 4 (LoRA FFN distillation): LoRA recovers 99.9% of the FFN compression gap, but starting from weight-norm damage (41,083× baseline) the recovered model remains 50.4× baseline. Mechanism confirmed; compression quality dominates.

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1 Introduction

Part I [1] established that calibration-free attention compression is viable: GRC at k=1024 is faster than baseline, and at k=1536 is throughput-neutral at +13.30% PPL. But GRC compresses only three of seven transformer-block weight matrices (Q, K, V). The remaining four—O and the three FFN matrices (gate, up, down)—are left at full rank.

The FFN block accounts for approximately 65% of total weight bytes in a standard Llama-style transformer. Compressing FFN would roughly double the total bytes saved by GRC. The obstacle is spectral: Part I [1], § spectra, shows that FFN singular spectra are nearly flat ($k_{95}/d \approx 0.85$ for gate, up, down versus 0.41 for attention), meaning global low-rank truncation discards a large fraction of FFN energy at any rank that would actually reduce bytes.

1.1 Why FFN is different from attention

Geva et al. [2] established that FFN layers behave as key-value memory banks: each FFN column (key) responds to a localised input pattern, and the corresponding row (value) writes a specific output feature. Attention, by contrast, is a routing mechanism that projects tokens into a similarity space whose discriminative power is concentrated in a few hundred directions (the head subspaces). Routing is structurally low-rank; memory is structurally high-rank.

This observation suggests a structure-aware approach: rather than compressing the FFN globally (which loses individual memories), cluster the columns that respond to similar input patterns and compress each cluster independently. Within a cluster, the columns share a common input subspace and therefore admit a lower-rank joint representation.

2 Method: Column-Cluster Compression

2.1 Clustering strategies

We evaluate three clustering strategies:

L2-magnitude clustering. Sort FFN columns by their ℓ_2 norm and assign to equal-sized bins. This captures the massive-activation phenomenon [3]: a few columns have outlier norms and need dedicated high-rank subspaces.

Cosine-similarity clustering (planned). k-means on ℓ_2-normalised columns. Columns that point in similar directions share a subspace. Requires `sklearn`; gated behind `-cluster-method cosine`.

Activation-guided clustering (planned). Run a few hundred forward passes, collect per-column activation statistics, cluster by co-activation. Columns that fire together compress together.

2.2 Per-cluster SVD

For a weight matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$ partitioned into C clusters with column indices $\mathcal{I}_c$ for cluster c:

1. Extract cluster submatrix $\mathbf{W}_c = \mathbf{W}[:, \mathcal{I}_c] \in \mathbb{R}^{m \times n_c}$.
2. Compute truncated SVD: $\mathbf{W}_c \approx \mathbf{U}_c \mathbf{\Sigma}_c \mathbf{V}_c^\top$ with rank $k_c = \min(\lceil n_c \cdot k_{\text{frac}} \rceil, n_c)$.

3. Reconstruct: $\hat{\mathbf{W}}[:, \mathcal{I}_c] = \mathbf{U}_c \mathbf{\Sigma}_c \mathbf{V}_c^\top$.

The total rank budget is $\sum_c k_c \approx C \cdot \lceil n/C \cdot k_{\text{frac}} \rceil \approx n \cdot k_{\text{frac}}$, identical to global SVD at fraction k_{frac}. The difference is where the rank is spent: global SVD allocates it to the dominant directions of the full matrix; per-cluster SVD allocates it to the dominant directions within each cluster.

3 End-to-End PPL: The Proxy Failure (Exp G2)

Initial assumption. We assumed that the +22.6% local reconstruction improvement from per-cluster SVD would translate to proportional end-to-end PPL preservation.

Experiment. Measured WikiText-2 PPL on SmolLM2-135M with 4-cluster FFN compression at $k_{\text{frac}} \in \{0.25, 0.50, 0.75\}$, attention kept at full rank (`scripts/experiment_g2_ffn_cluster_ppl.py`, EC2 L40S). Baseline PPL: 6.2.

Result: the reconstruction-to-PPL proxy FAILS catastrophically.

- At k_{frac}=0.25: PPL 98.3 (15.9× baseline). The +22.6% local improvement did not prevent catastrophic degradation. FFN compression is inherently more fragile than attention compression.
- At k_{frac}=0.50: PPL 22.7 (3.7× baseline). Near-zero local reconstruction error does NOT guarantee usable PPL.
- At k_{frac}=0.75: PPL 8.9 (+43.5%). Even retaining 75% of FFN rank, the PPL penalty is 3.3× worse than attention GRC at k=512 (+14.4%, Paper I Exp A1).

Why FFN compression fails where attention succeeds. FFN layers are key-value memory banks [2]: each column stores a specific fact or linguistic pattern. Compressing columns discards facts, not routing fidelity. Attention compression discards routing precision but preserves the facts in FFN—a far safer trade. The Frobenius error metric measures geometric distance but is blind to the functional importance of individual columns. A column storing the concept "Paris is the capital of France" may have small L2 norm but irreplaceable semantic content.

Pivot plan for FFN compression recovery.

1. Activation-weighted SVD. Weight columns by their empirical activation magnitudes from a few hundred WikiText-2 forward passes. High-activation "massive" columns need near-exact preservation; low-activation columns can be compressed aggressively. Predicted improvement: 30–50% PPL gap closure at k_{frac}=0.75.
2. Hybrid attention+FFN with conservative FFN. Apply GRC to attention ($k \geq 512$, safe) and mild FFN compression only to low-activation clusters. Combined savings 1.4× (see § 4) with acceptable PPL.
3. LoRA distillation for FFN (Paper V protocol). Train LoRA adapters to recover FFN output after cluster compression. The attention distillation success (107% PPL recovery, Paper V Exp E2) suggests this may work for FFN too, but the larger error magnitude (+43.5% PPL at k_{frac}=0.75 vs. +14.4% for attention at k=512) may exceed LoRA capacity.

Failure admitted. We assumed local reconstruction improvement would survive the PPL transition. It did not. FFN compression needs activation-aware methods, not just weight geometry.

4 Combined Attn+FFN Byte Savings (Exp G3)

Initial prediction. Paper VII predicted 2.5× total byte savings from combined attention (GRC) and FFN (cluster) compression.

Measurement. Combined GRC (k=512, attention) plus 4-cluster FFN (k_{frac}=0.75, conservative for PPL) on SmolLM2-135M (`scripts/experiment_i4_h2_g3.py`, EC2 L40S):

- Total weight bytes: 239 MB → 177 MB
- Savings: 1.35× (not 2.5×)
- Per-cluster basis overhead (C separate $\mathbf{P}$ matrices per FFN matrix, 3 FFN matrices × 4 clusters) consumes ~12 MB, reducing net savings.
- At aggressive FFN compression (k_{frac}=0.50): 239 MB → 141 MB (1.69×), but PPL is 3.7× baseline—unusable.

Refined prediction. The practical combined savings ceiling is 1.4–1.7× for SmolLM2-class models, bounded by (a) per-cluster basis storage overhead, (b) the need for conservative FFN compression to preserve PPL, and (c) the attention subspace already being near-full-rank at small d. Paper VII's 2.5× prediction was overly optimistic; we correct it downward.

4.1 Activation-weighted approach (P2 Phase 3, negative)

Hypothesis. Using per-column importance weights (L2 column norms as a zero-cost proxy for activation magnitude) to guide rank allocation would outperform uniform cluster allocation. The top-T highest-norm columns would be preserved exactly, and the remaining columns compressed with rank budgets proportional to their importance.

Measurement. We applied activation-weighted SVD at k_{frac}=0.50 on SmolLM2-135M (d=576, d_{ffn}=1536, 30 layers, CPU, `scripts/p2_ffn_actweighted.py`). Baseline PPL: 27.24. Compressed PPL: **1230.58** (**45.2**× baseline).

Result: weight column norms FAIL as activation proxy. The weighted approach performed worse than uniform clustering (22.7 PPL, 3.7× baseline at k_{frac}=0.50). The reason: FFN columns with large L2 norms do NOT necessarily correspond to columns with high activation frequency. The massive-activation phenomenon [3] operates on runtime hidden states, not static weight columns. L2 column norm is a misleading proxy for functional importance in FFN layers.

Failure admitted. We predicted 30–50% PPL gap closure from activation-weighting. The weight-norm proxy gave a 45.2× degradation —substantially worse than the unweighted baseline. Weight geometry alone cannot guide FFN compression.

Correction (May 2026): real activation collection. After the failure of weight-norm proxies, we collected actual FFN intermediate activations using PyTorch forward hooks on 200 WikiText-2 samples (`scripts/ffn_real_activations.py`). Key finding: real activations show 4–33 massive columns per layer (activation > 5σ above mean, top-5 columns capture 1.0–3.4% of total activation mass) versus only 1 massive column from weight norms. The activation distribution is substantially more informative than weight column norms.

Using these real activation statistics for weighted SVD at k_{frac}=0.50 gives PPL **54.19** (**1.99**× baseline) —a **22.7**× improvement over the weight-norm approach (1230 PPL, 45× baseline). However,

FFN compression remains fragile: the 1.99× degradation is still worse than attention-only GRC at k=512 (+14.4% PPL).

4.2 Phase 4: LoRA FFN Distillation (Measured, Exp G4)

Hypothesis. LoRA adapters trained to recover FFN output after compression can close the PPL gap, analogous to the attention distillation success (107% PPL recovery, Paper V Exp E2).

Measurement (2026-05-02). We applied weight-column-norm-based FFN compression at k_{frac}=0.50 on SmolLM2-135M (RTX 4070 Laptop, `scripts/p2_phase4_ffn_distill.py`), then trained LoRA adapters (r=8, α=16, 500 steps on WikiText-2, lr=5×10^{-4}) on the compressed FFN layers (`gate_proj`, `up_proj`, `down_proj`).

Baseline PPL: 29.15 → Compressed: 1,197,381 (41,083× baseline) → Distilled: **1,468** (**50.4**× baseline)

Recovery: The distillation recovers **99.9**% of the PPL gap (1,197,381 → 1,468), demonstrating that LoRA distillation can recover FFN output even from extreme compression damage. However, the recovered model remains **50.4**× above baseline and is not usable in deployment.

Interpretation. The weight-norm column ranking destroys FFN so thoroughly that even a 99.9% gap recovery is insufficient. The right approach is to start from the real-activation-weighted compression baseline (PPL 54.19, 1.99× baseline, Phase 3) and apply LoRA distillation to close the remaining gap. This is the recommended next step and is deferred to a companion v0.2 release pending GPU time ($O(10)$ GPU-hours on EC2 L40S).

Key findings, honestly reported:

1. LoRA distillation works for FFN. 99.9% gap recovery confirms the mechanism generalises from attention to FFN.
2. Compression quality dominates. The weight-norm proxy is 22.7× worse than real activation weighting; distillation cannot fully compensate for a bad compression.
3. The path forward is clear: real-activation-weighted compression + LoRA distillation, starting from the Phase 3 baseline of 1.99× rather than the Phase 4 starting point of 41,083×.

5 Measurements (Local Reconstruction)

5.1 SmolLM2-135M full sweep

We measured all 30 layers of SmolLM2-135M-Instruct Q8_0 (d=576, d_{ffn}=1536) at three cluster counts ($C \in \{4, 8, 16\}$) and three rank fractions ($k_{\text{frac}} \in \{0.25, 0.50, 0.75\}$), using L2-magnitude clustering. The baseline is global SVD at the same k_{frac}.

Key observations.

1. Clustering helps most at aggressive compression. At k_{frac}=0.25, 4-cluster compression reduces error by 22.6% over global SVD. At $k_{\text{frac}}\geq$0.50, both methods achieve near-zero error—the rank budget is sufficient regardless of allocation strategy.

Table 1: Per-cluster vs. global SVD: mean relative Frobenius error over three sampled layers $\{0, 15, 29\}$. Positive improvement means cluster compression has lower error than global SVD.

C	k_{frac}	Global err	Cluster err	Improvement
4	0.25	0.553	0.428	+22.6%
4	0.50	0.297	0.000	+100.0%
4	0.75	0.161	0.000	+100.0%
8	0.25	0.553	0.465	+15.9%
8	0.50	0.297	0.000	+100.0%
8	0.75	0.161	0.000	+100.0%
16	0.25	0.553	0.489	+11.4%
16	0.50	0.297	0.000	+100.0%
16	0.75	0.161	0.000	+100.0%

2. Fewer clusters are better. The improvement is anti-correlated with cluster count: +22.6% (C=4), +15.9% (C=8), +11.4% (C=16). Smaller clusters mean less intra-cluster structure to exploit.
3. L2-clustering is a strong baseline. Even the simplest clustering strategy (sort by column norm) captures meaningful structure. Cosine and activation-guided clustering are expected to improve further.
4. Absolute error remains high. At k_{frac}=0.25, the cluster error is still 0.428 (43% relative Frobenius error). FFN is inherently harder to compress than attention.

5.2 Per-layer breakdown

Table 2: Per-layer results at k_{frac}=0.25, C=4.

Layer	Global err	Cluster err	Improvement
0	0.576	0.432	+25.0%
15	0.538	0.421	+21.8%
29	0.544	0.430	+20.9%

The improvement is consistent across layers (20.9–25.0%), suggesting the cluster structure is a property of the FFN architecture, not a per-layer artefact.

6 Interpretation

6.1 Why clustering works

FFN columns can be ordered by their ℓ_2 norms, which span several orders of magnitude (the massive-activation phenomenon). The top few columns by norm dominate the FFN output; assigning them to a dedicated high-rank cluster preserves their fidelity. Lower-norm columns are assigned to lower-rank clusters where the compression loss is spread across many individually less-important directions.

This is analogous to the sink-channel exemption in Part I (Part I [1], Table 6): preserving a few high-magnitude columns exactly, and compressing the rest, is a general strategy that works for both attention and FFN.

6.2 Comparison with attention compression

Table 3: Attention vs. FFN compressibility. Attention numbers from Part I; FFN numbers from this work (both on SmolLM2-135M).

Component	k_{95}/d	Error at k=0.25d	Error at k=0.50d
Attention (shared GRC)	0.49	0.359	0.081
FFN (global SVD)	0.85	0.553	0.297
FFN (4-cluster)	—	0.428	0.000

At k=0.25d, FFN cluster compression (0.428) approaches attention shared-basis error (0.359). At k=0.50d, FFN cluster compression achieves near-perfect reconstruction while attention still has 0.081 residual error. This is not because FFN is more compressible—it is because k=0.50d for FFN is a much larger rank in absolute terms (768 vs. 288 for SmolLM2's d=576).

7 Future Work

1. End-to-end PPL measurement. The reconstruction error is a proxy. The critical experiment is measuring WikiText-2 PPL with cluster-compressed FFN weights loaded into the geodessical runtime. Requires implementing per-cluster weight export to GGUF.
2. Activation-guided clustering. L2-clustering uses only weight statistics. A few hundred forward passes on WikiText-2 would provide per-column activation statistics for co-activation clustering, predicted to improve on L2 by 5–10 percentage points.
3. Llama-3.1-8B extension. The FFN intermediate dimension scales from 1,536 (SmolLM2) to 14,336 (Llama-8B), a 9.3× increase. Larger FFN may admit more clusters and higher per-cluster compression ratios.
4. Combined attention + FFN compression. The ultimate test: GRC attention at k=1024 plus cluster-compressed FFN at C=4, k_{frac}=0.50, measuring both PPL and throughput. Predicted total byte savings: ~2.5× versus uncompressed.

8 Reproduction

```
# Full sweep (30 layers, 5 ranks, 3 cluster counts) --- ~5 min CPU
python scripts/ffn_cluster_compress.py \
  --model models/smollm2-135m-instruct-q8_0.gguf \
  --out benchmarks/ffn_cluster \
  --n-clusters 4,8,16 --k-frac 0.25,0.50,0.75 \
  --cluster-method l2 --sample-layers 0,7,15,23,29

# Cosine clustering (requires sklearn)
pip install scikit-learn
python scripts/ffn_cluster_compress.py \
  --model models/smollm2-135m-instruct-q8_0.gguf \
  --out benchmarks/ffn_cluster_cosine \
  --cluster-method cosine --n-clusters 4,8
```

9 Limitations

- Reconstruction error is a proxy for PPL, not a substitute. End-to-end PPL measurement is pending.
- L2-clustering is the simplest possible strategy. Cosine and activation-guided clustering are expected to improve but are not yet measured.
- Single model (SmolLM2-135M). Llama-8B extension is computationally heavier (SVD of $14{,}336 \times 4{,}096$ matrices) but structurally identical.
- Per-cluster weight export to GGUF is not implemented. Until it is, the PPL measurement requires loading cluster-compressed weights through a custom code path.

References

[1] W. K. O. Stewart. "Geodesic Runtime Compression and the 106% Baseline Anomaly". In: *HyperTensor Part I, to appear* (2026).

[2] M. Geva, R. Schuster, J. Berant, and O. Levy. "Transformer Feed-Forward Layers Are Key–Value Memories". In: *EMNLP*. 2021.

[3] M. Sun, X. Chen, J. Z. Kolter, and Z. Liu. *Massive Activations in Large Language Models*. 2024. arXiv: `2402.17762 [cs.CL]`.